

JULIAN D. SANTAMARIA P

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EDUCATION

Master of Science in Engineering (Student)

Universidad de Antioquia, Medellin, Colombia

Jan 2024 - Dic 2025

Bachelor of Science in Electronics Engineering (GPA 4.16/5)

Universidad de Antioquia, Medellin, Colombia

Feb 2018 - Dic 2023

ACADEMIC EXPERIENCE

Intern

Télécom Paris, Institut Polytechnique de Paris, France.

Research on domain adaptation in computer vision for biodiversity monitoring.

Supervised by Jhony H. Giraldo.

Jan 2025 – Jun 2025

Junior Researcher

SISTEMIC Research Group, Universidad de Antioquia, Colombia.

Research on bridging the gap between foundation models and species recognition.

SISTEMIC Research Group, Universidad de Antioquia, Colombia.

Machine learning for monitoring the critically endangered species *Atlapetes blancae*.

Jan 2024 – Present

Jan 2022 – Dec 2023

Teaching Assistant

Universidad de Antioquia, Medellín, Colombia.

Support for the undergraduate course “Computer Science I”.

Universidad de Antioquia, Medellín, Colombia.

Academic tutoring in the course “Fundamentals of Digital Electronics”.

Jan 2024 – Present

Jan 2021 – Dec 2023

Visiting Research Scholar

CinfonIA Research Group, Universidad de los Andes, Colombia.

Research on wildlife image segmentation.

Supervised by Pablo Arbelaez.

Jul 2025 – Aug 2025

RESEARCH INTERESTS

Out-of-Distribution Generalization, Domain Adaptation, Wildlife Monitoring, Multimodal Learning, Graph Neural Networks, Computer Vision, Machine Learning.

PUBLICATIONS

- **Julian D. Santamaria**, Claudia Isaza, Jhony H. Giraldo, “*CATALOG: A Camera Trap Language-guided Contrastive Learning Model*”, **Oral Presentation** (Around 8% of all submitted papers) at IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025, a Rank A conference. 
- **Julian D. Santamaria P**, Jhony H. Giraldo, Angelica Diaz-Pulido, Claudia Isaza “*Audio vs. Visual Approach to Monitor the Critically Endangered Species *Atlapetes blancae*: Developing Deep Learning Models with Limited Data*”, The 2024 IARIA Annual Congress on Frontiers in Science, Technology, Services, and Applications, 2024. 

PAPERS UNDER REVIEW

- **Julian D. Santamaria**, Claudia Isaza, Jhony H. Giraldo. “*WildIng: A Wildlife Image Invariant Representation Model for Geographical Domain Shift*”. Submitted to the International Journal of Computer Vision (IJCV).

POSTER PRESENTATIONS

- **Audio and Video Analysis using Machine Learning for Monitoring the critically endangered species *Atlapetes blancae***, International Conference on Computer Vision - ICCV 2023 - LatinX Workshop. 2023.

FELLOWSHIPS AND SCHOLARSHIPS

- **NVIDIA Academic Grant**, NVIDIA AI *Bridging the Gap between Foundation Models and Camera Trap Image Recognition*. **Role:** Recipient. **10,000 A100 GPU-Hours**. Awarded for research proposal co-written with my advisor **Jhony H. Giraldo** Jan 2025 - Jun 2025
- **Estudiante Instructor Scholarship**, Universidad de Antioquia *Funding for the Master in Engineering to support the thesis “Integración de modelos de fundación para la identificación automática de animales.”* **Role:** Recipient. **€20,450**. 2024 - 2025.
- **CV Foundation Travel Grant**, International Conference on Computer Vision (ICCV) *Scholarship to support travel and attendance at ICCV 2023*. **Role:** Recipient. **€1,000**. 2023
- **LatinX in AI Travel Scholarship**, LatinX in AI *Grant awarded to support presentation in the LatinX in AI Workshop at ICCV 2023*. **Role:** Recipient. **€400**. 2023
- **Colombian Government Undergraduate Scholarship**, Ministerio de Educación Nacional *Funding for undergraduate studies in Electronic Engineering*. **Role:** Recipient. **€17,296**. 2018 - 2023.

AWARDS AND HONORS

- **Academic Excellence Award**, Universidad de Antioquia *Awarded to the highest-performing first-level student in the Electronics Engineering program*. **Semester: 2018-1**.
- **Academic Excellence Award**, Universidad de Antioquia *Awarded to the highest-performing eighth-level student in the Electronics Engineering program*. **Semester: 2023-1**.

PROFESSIONAL SERVICE

Journal Reviewer: IEEE Transactions on Biomedical Engineering and EURASIP Journal on Image and Video Processing.

COMPUTER SKILLS

Programming Languages

Python, Matlab, C/C++.

Relevant Libraries

PyTorch, PyTorch Geometric, OpenCV, Transformers (HuggingFace).

Tools & Frameworks

Weights & Biases (wandb), Git, L^AT_EX, Inkscape.

LANGUAGES

Spanish: Native Language.

English: B2. IELTS: 5.5/9 (2017), Professional Working Proficiency.